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Module Code: COMP50008

Module Name: Enterprise Cloud and Infrastructure Automation-2

Abstract

NexaCart Solutions (Pvt) Ltd, a Colombo-based digital commerce company processing about 60,000 orders monthly, encounter significant challenges with its on-premises infrastructure, such as manual provisioning, single points of failure, limited disaster recovery, and insufficient scalability during traffic spikes. This report details the research, design, and partial implementation of an AWS cloud solution, including the 7Rs migration framework, AWS Well-Architected Framework, infrastructure analysis, CloudFormation deployment, and disaster recovery planning for NexaCart's cloud migration.

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1. Mitigation Framework & AWS Best Practices

1.1 What is Cloud Mitigation

Cloud Mitigation is the process of moving applications, data, workloads, and IT infrastructure from a company's private on-premises data centers to a cloud-based infrastructure, such as private, hybrid, or public (Microsoft, 2026).
By choosing to migrate to using cloud resources, enterprises benefit from cost savings, greater levels of scalability, improved disaster recovery, improved business continuity, and new technologies such as AI/ML.

Often, organizations find themselves migrating due to old legacies in terms of their physical infrastructure or the increasing cost of running their local data centers that support numerous employees who work remotely or work from home. Some companies also migrate to accelerate the delivery of new digital solutions to their customers through innovation/evolution speed, while others are simply looking for more efficient ways to manage their own team data storage needs or to take advantage of the cloud's native capabilities that aren't available on premises (IBM, 2025).

1.2 Types of Cloud Mitigation

Cloud migration types come in different forms depending on where applications are moving and why. They include the following categories:

- **Data center migration** - Moving your entire, physical data center (servers, storage, networking) to a new, cloud-based environment.
- **Hybrid cloud migration** - Integrating an existing on-premises infrastructure with a public cloud. This may include migrating certain workloads to the cloud while retaining sensitive/legacy systems on premises.
- **Cloud-to-cloud migration** - Transferring applications, data, and existing workloads between two cloud service providers, such as moving from AWS to Azure.
- **Multicloud migration** - Having your applications, data, and services hosted by multiple public cloud providers at the same time, usually done for reasons such as reducing vendor lock-in and/or taking advantage of the best features from each provider.
- **Workload-specific migration** - Moving one specific, targeted application, service, or database to the cloud, rather than migrating your entire IT infrastructure all at once.

1.3 Managed service models

IaaS

IaaS allows you to quickly scale up and down based on your needs and charges you only for what you use. IaaS allows you to avoid the complexities and costs of purchasing and managing your own physical servers or datacenter-related infrastructure.

PaaS

PaaS provides you with the same advantages of IaaS, but also provides you with managed developer tools, databases, analytics, and operating systems. PaaS can decrease your development costs and speed your time to market.

SaaS

SaaS provides you with the same things you get with IaaS and PaaS, but it also provides you with the convenience of app hosting that allows the end user to connect to their cloud-based application, such as e-mail and other productivity applications, over the internet.

1.4 Cloud migration stages

There are different technical processes and best practices for migrating different workloads and creating different cloud configurations, but all cloud migrations follow the same basic six stages:

- **Define strategy** - Define the business justification for migration and the expected outcomes.
- **Plan** - Align the cloud migration plan to the desired business outcomes.
- **Ready** - Prepare the on-premises and cloud environments for the migration.
- **Migration** - Move the workloads to the cloud environment.
- **Govern** - Benchmark and implement security and cost governance best practices for the cloud environment and workloads.
- **Manage** - Use proven methodologies for ongoing management of the cloud solution.

1.5 Common cloud migration benefits

- **Optimized costs -** In addition to being able to optimize workloads for cost, there are significant savings in equipment, maintenance, and real estate costs.
- **Flexibility and scalability** - Get exactly the right computing resources when you need them on demand and never pay for equipment that you don't use. And with some cloud platforms' global presence, you can run workloads closer to where your customers are for better performance.

- **Enhanced security** - Public clouds make it easier to help protect your data and apps by providing built-in security at the platform level, as well as a variety of specialized security tools. In many cases, security patching is done automatically by the cloud provider.
- **Compliance** - For highly regulated industries like finance, healthcare, and government, some cloud platforms have specialized offerings to meet compliance requirements more easily. Backup, recovery, and failover. Most cloud providers offer built-in and one-click backup recovery capabilities, including the ability to store backups in different geographic regions.
- **Simplified management and monitoring** - If the cloud provider offers a central management tool, it's easy to manage and monitor datacenter and cloud resources from a single screen.

1.6 Challenges of cloud migration

When transitioning to cloud-based services, there are many challenges, and therefore, it is extremely important to implement a well-structured strategy. Legacy systems normally contain many complex, undocumented dependencies. Consequently, it is often difficult to ascertain the full scope of the potential impact associated with migrating legacy systems to the new cloud environment. Performance issues can arise when applications developed specifically for an on-premises infrastructure do not perform optimally within a cloud environment. Additionally, applications that rely upon real-time data processing or ongoing cross-communication between their various components may suffer from delays in data transmission due to network latency.

Numerous security and compliance concerns need to be carefully considered before moving to a cloud-based solution to ensure that your data is protected and that meets all regulatory requirements. Skills gaps are a major obstacle, and 58% of decision-makers worldwide indicated that they do not have skilled employees who can help them operate in the cloud. The cost of operating in the cloud will continue to increase if the organization does not have a governance structure in place that defines how its teams should make decisions about the use of cloud resources (IBM, 2025).

1.7 The 7Rs of Cloud Mitigation

The 7Rs Model is the method for determining how to migrate each of your systems to the Cloud. It started out as a 5Rs Model, and then AWS added two more. The idea of using the 7Rs Model is to demonstrate that not all workloads will be migrated the same way; therefore, each workload requires an individual Migration Plan for determining Cost, Effort, and Long Term Value, as well as AWS Best Practices for successful migration (NetApp, 2026).

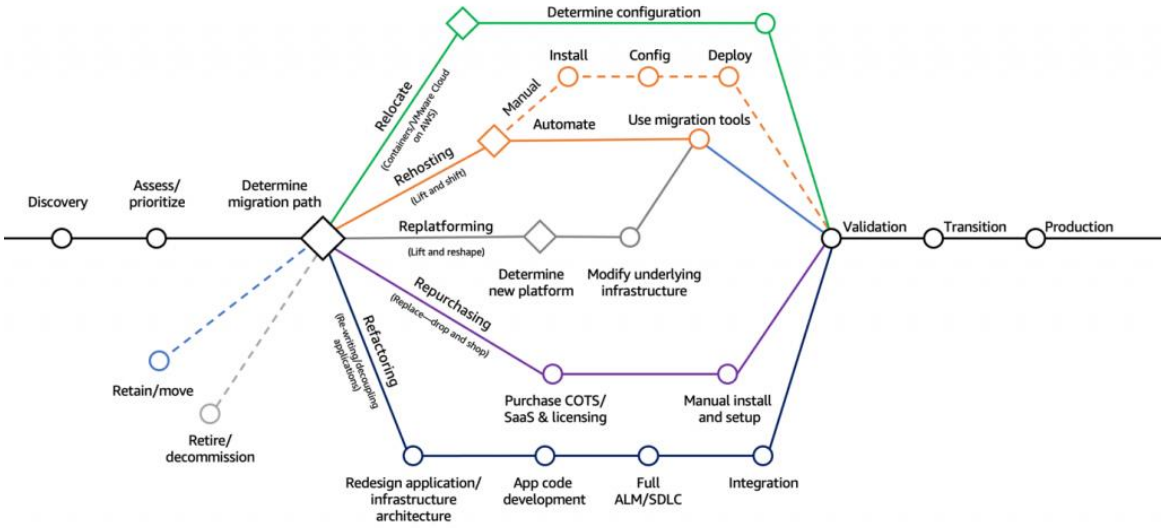


Figure 1:7 AWS Strategies for Migrating to the Cloud.

1.7.1 Rehost – Lift and Shift

Rehosting refers to the migration of an application and any related dependencies from on-premises infrastructure to the Cloud without any application changes. It is as if you are lifting the entire current environment and simply moving it to cloud compute resources (AWS, 2026).

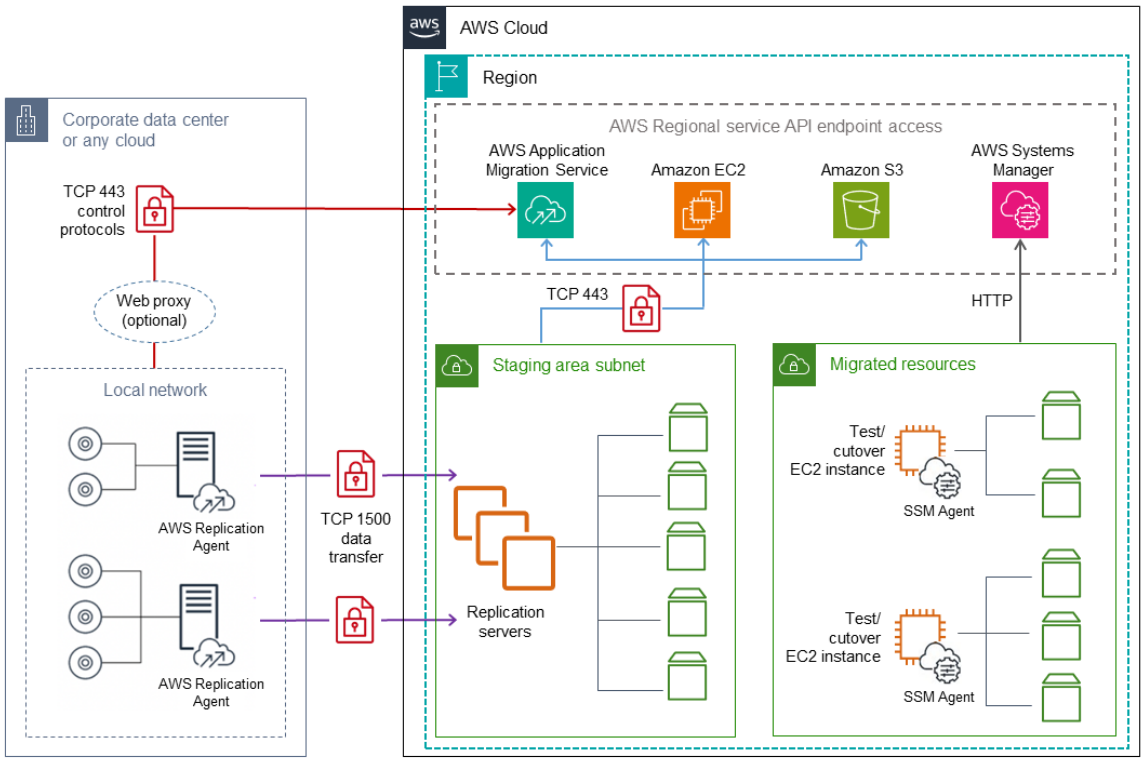


Figure 2: Rehost on-premises workloads in the AWS Cloud.

The above diagram illustrates rehosting (discovering and migrating servers from an on-premises source environment to AWS) by using Cloud Migration Factory and AWS Application Migration Service.

Advantages:

- Fastest migration approach with minimal disruption to existing workflows.
- No application code changes required.
- Reduces data centre overhead immediately.
- Fully automated by AWS Application Migration Service (MGN).

Possible Risks and Overhead:

- Does not leverage cloud-native optimisations.
- Operating costs may remain high if instances match on-premises specifications.
- No architectural improvements; technical debt persists.
- Estimated cost reduction of 20–30% compared to on-premises, but less than cloud-native approaches.

NexaCart Application: Web servers (3× Dell PowerEdge T40) can be rehosted to EC2 instances as an initial step for rapid migration, with further optimisation planned.

1.7.2 Replatform (Lift, Tinker, and Shift)

Replatforming is the act of cloud-optimising certain parts of your existing system (without changing all of its underlying architecture) onto a cloud platform (for example, converting your self-managed database to run on Amazon RDS, or putting your existing application server behind an Elastic Load Balancer) (AWS, 2026).

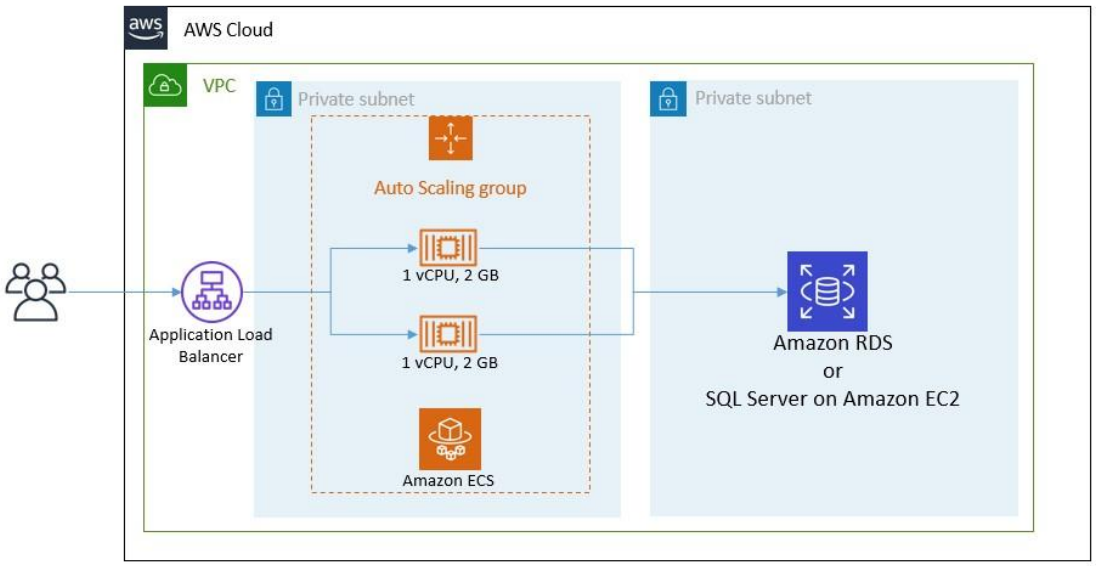


Figure 3: MvcMusicStore architecture on Amazon Elastic Container Service.

Advantages:

- Moderate performance and operational gain (without re-architecting completely).
- Reduced administrative overhead (AWS handles patching, backups for RDS, etc).
- A better cost optimisation compared to pure rehosting.
- You will gain some of the benefits of being a cloud-native organisation (managed services, readiness for automated scale, etc).

Possible Risks and Overhead:

- You will need to make application-level changes (changing connection strings/configurations).
- You will have a slightly longer time frame for migration compared to pure rehosting.
- You will incur a moderate one-time labour cost for the migration.

NexaCart Application: The application server, database server, and file server are suitable for replatforming by migrating to EC2 with ALB, Amazon RDS Multi-AZ, and Amazon S3, respectively.

1.7.3 Repurchase (Drop & Shop)

Repurchasing replaces an existing application with a different product, typically a SaaS solution. For example, migrating from a self-hosted email server to Microsoft 365 or Amazon WorkMail (AWS, 2026).

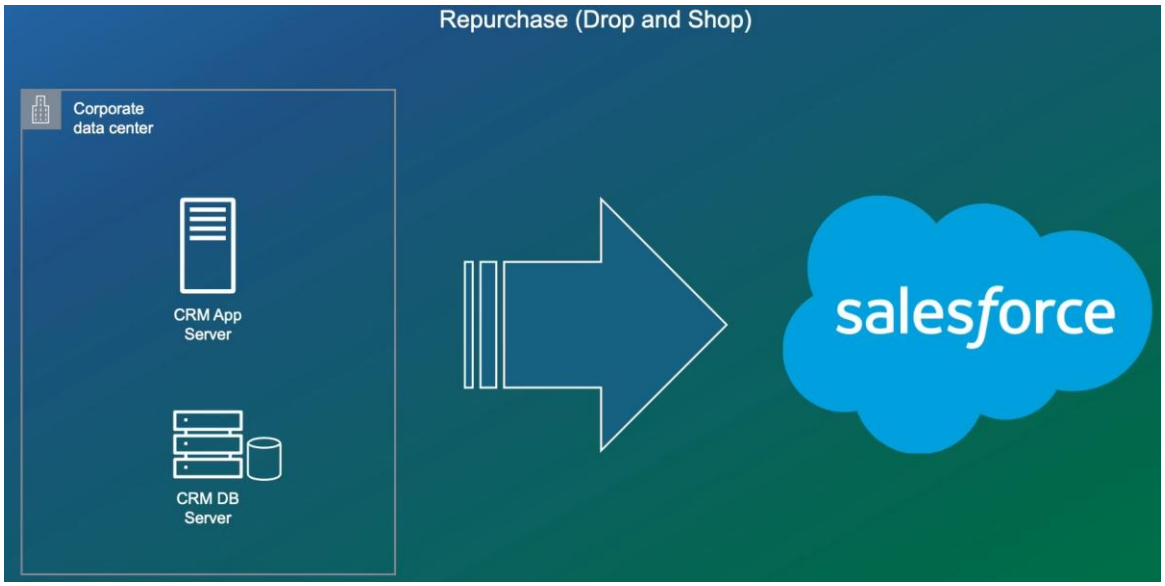


Figure 4: Cloud Mitigation via Repurchase.

Advantages:

- You will always have access to the latest version.
- You can help reduce your IT operational costs.
- You can have predictable subscription billing.

Possible Risks and Overhead:

- You may need to pay for new licenses (SaaS subscriptions).
- Data migration may be complex.
- Users may have to learn how to use the new application
- You may lose some customisation in some instances.

A great example of an application that should be repurchased is Microsoft Exchange Server 2019 (on-premises). By replacing this application with either Amazon WorkMail or Microsoft 365, you will eliminate email infrastructure from your business.

1.7.4 Re-architecture/ Refactor

This term refers to redeveloping an application into a cloud-native app by moving to a micro-service architecture, serverless (AWS Lambda), containerising (Amazon ECS/EKS), or using an event-driven based-method (AWS, 2026).

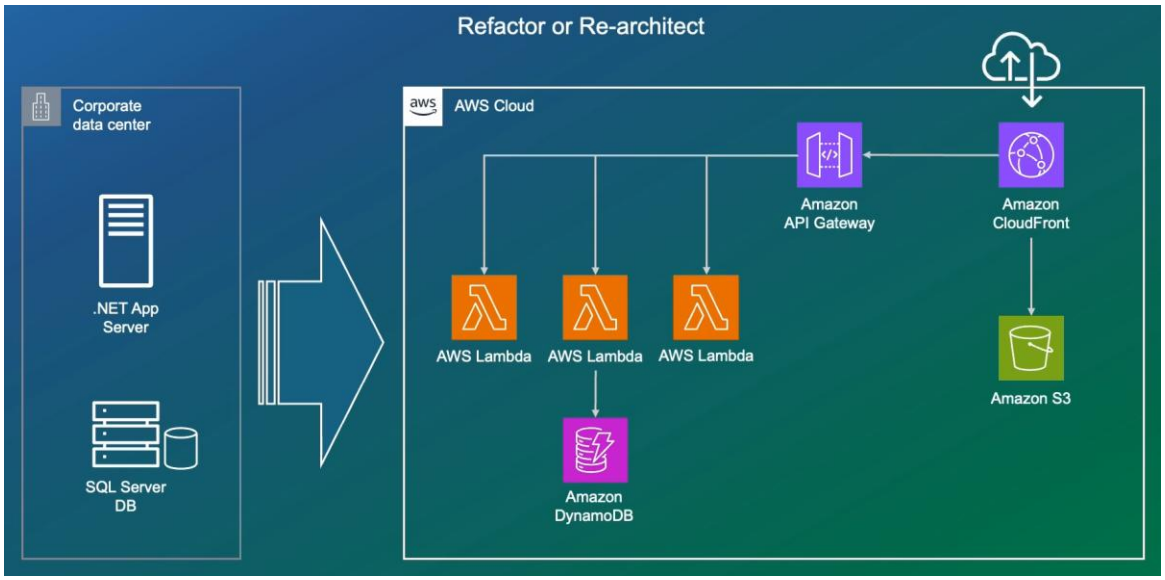


Figure 5: Cloud Mitigation via Refactor.

Advantages:

- Maximum cost-efficiency and efficiency over time.
- Full cloud elasticity and expandability.
- Enables continuous deployment and DevOps maturity.
- Ideal for applications requiring significant feature velocity.

Possible Risks and Overhead:

- Highest upfront development investment
- Longest timeline to production
- Requires strong cloud engineering expertise
- Risk of scope creep during re-architecture

NexaCart Application: The long-term goal is to refactor order management and inventory APIs into Lambda-based microservices behind API Gateway, supporting planned AI-based demand forecasting and live tracking features.

1.7.5 Retire

Retiring means identifying and decommissioning applications or infrastructure components that are unnecessary, redundant, or lack sufficient business value to justify migration costs.

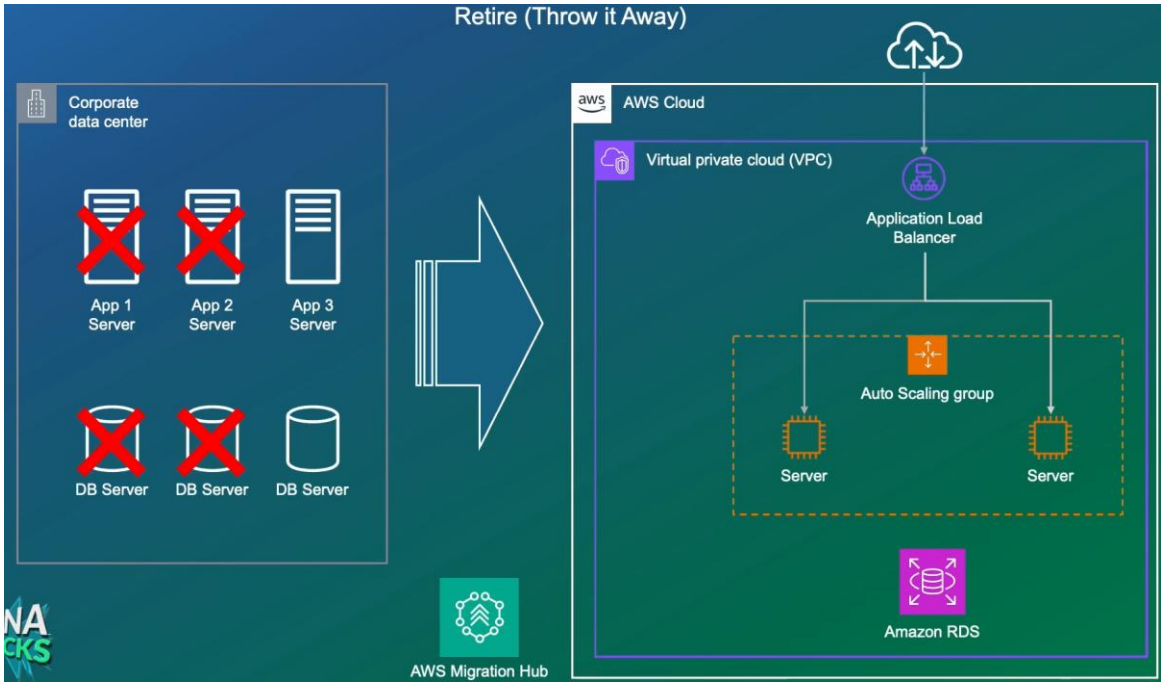


Figure 6: Cloud Mitigation via Retire.

Advantages:

- Reduces migration scope and cost.
- Eliminates security risk from orphaned systems.
- Simplifies the overall cloud architecture.

Possible Risks and Overhead:

- Accidentally removing systems still used by hidden integrations.
- Deleting data needed for audits or compliance.
- Upsetting users who still depend on "legacy" tools.

NexaCart Application: By retiring the old legacy dev/test virtualised server (8 vCPU, 16GB RAM), thus relocating an entire set of duplicate test environments to an on-demand model using EC2 spot-instance, it will not only reduce ongoing compute costs but also allow more flexibility in scaling our testing environments for additional needs.

1.7.6 Retain

Retaining involves deliberately keeping certain workloads on-premises due to regulatory compliance, technical complexity, recent hardware investment, or dependencies that make short-term migration impractical.

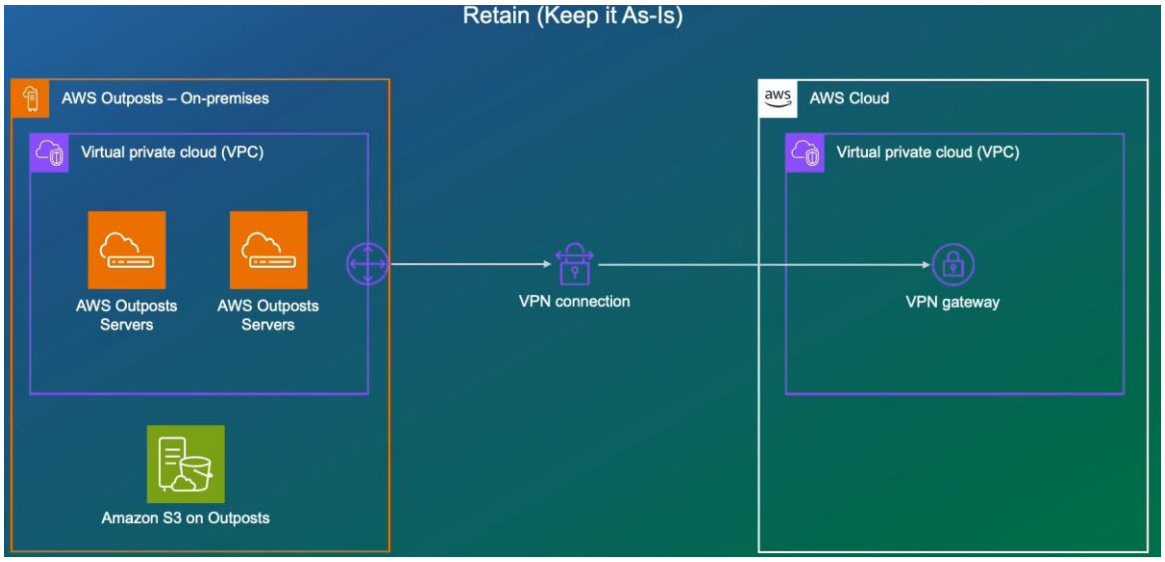


Figure 7: Cloud mitigation via Retain

Advantages:

- Manages migration risk by limiting scope
- Allows time for compliance assessment
- Protects recent hardware investments

Possible Risks and Overhead:

- Extra complexity running both on-prem and cloud.
- Latency issues between retained systems and AWS workloads.
- On-prem security and patching may lag behind the cloud.

NexaCart Application: Workstations (20× Dell OptiPlex PCs) and local LAN switching infrastructure remain on-premises to support office operations and do not require cloud hosting.

1.7.7 Relocate

Relocating is a hypervisor-level lift-and-shift, moving virtual machines from an on-premises VMware environment to VMware Cloud on AWS, or between AWS regions or accounts, without changing the OS or application layer.

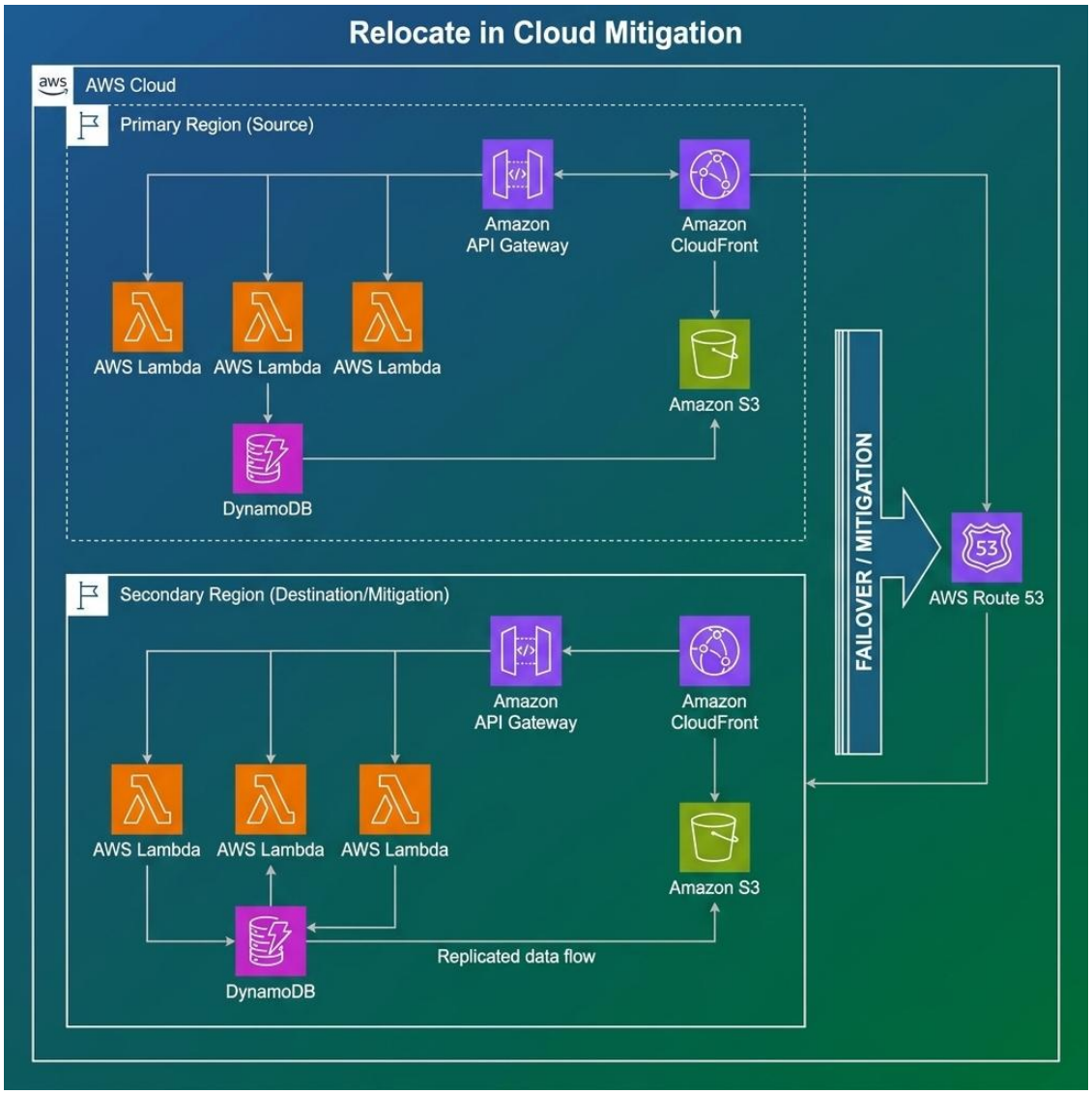


Figure 8: Cloud mitigation via Relocate

Advantages:

- Extremely fast migration for VMware workloads.
- No changes to existing VMs.
- Useful for data sovereignty or account restructuring.

Possible Risks and Overhead:

- Lifting existing performance and security issues into AWS unchanged.
- Higher long-term cost vs modernising to native AWS services.

- Extra skills and tooling needed to run both VMware and native AWS.

NexaCart Application: If NexaCart used VMware-based VMs for dev/test, the Relocate strategy via VMware Cloud on AWS would enable rapid migration. This is currently less applicable due to the bare-metal server environment.

1.8 AWS Well-Architected Framework

The AWS Well-Architected Framework provides customers and partners with a consistent method for evaluating their architecture and designing scalable solutions. The framework includes six main factors to consider.

1.8.1 Pillar 1: Operational Excellence

The definition of operational excellence includes giving value to a company through the operation of its systems, as well as monitoring those systems for continuous improvement of both their operations and monitoring processes (AWS, 2026). The underlying philosophy of operational excellence is to treat operations like source code (also known as infrastructure as source code (using CloudFormation) and use continuous integration/continuous deployment (CI/CD) of run books to automate operations). The goal should be to improve an organization's operations by making small, frequent, reversible changes using CI/CD pipelines to update the run book.

Design Principles:

- Perform operations as code (Infrastructure as Code via CloudFormation)
- Organize teams around business outcomes
- Implement observability for actionable insights
- Safely automate where possible
- Make small, frequent, reversible changes
- Refine operations procedures frequently
- Anticipate failure and run pre-production game days
- Learn from all operational failures
- Use managed services

NexaCart is violating this particular principle through its current workflow of manually provisioning (configuring each server by hand). Transitioning their operation toward using CloudFormation templates, AWS Systems Manager to patch their servers, and creating CloudWatch dashboards for visibility will provide for proactive instead of reactive operations. A specific instance of the NexaCart-advanced-ASG CloudFormation stack

implementing this principle can be seen in the full codification of the entire infrastructure as a reusable CloudFormation template.

1.8.2 Pillar 2: Security

The Security pillar describes how to take advantage of cloud technologies to protect data, systems, and assets - covering identity, detection, infrastructure protection, data protection, and incident response (AWS, 2026).

Design Principles:

- Implement a strong identity foundation
- Maintain traceability
- Apply the principle of least privilege via IAM
- Enable traceability through AWS CloudTrail logging
- Automate security best practices
- Protect data in transit (TLS) and at rest (KMS encryption)
- Keep people away from data
- Prepare for security events

Application to NexaCart: NexaCart handles customer payment data and personal information, making this pillar critical. The deployed CloudFormation stack creates tiered Security Groups (ALBSecurityGroup, WebSecurityGroup, DBSecurityGroup) enforcing network-level micro-segmentation. Amazon GuardDuty provides continuous threat detection, while AWS WAF on the ALB protects against common web exploits - essential for PCI-DSS compliance around payment processing.

1.8.3 Pillar 3: Reliability

Reliability ensures a workload performs its intended function correctly and consistently when expected, covering automatic recovery from failures, horizontal scaling, and disaster recovery (AWS, 2026).

Design Principles:

- Automatically recover from failure
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity — use Auto Scaling
- Manage change through automation

Application to NexaCart: NexaCart's single application server and database server represent catastrophic single points of failure. The AWS architecture resolves this through Multi-AZ RDS deployment (automatic failover in 60–120 seconds), EC2 Auto Scaling Groups spanning two Availability Zones (NexaCart-Web-ASG, min:2, max:4), and an Application Load Balancer distributing traffic across healthy instances.

1.8.4 Pillar 4: Performance Efficiency

Performance Efficiency focuses on using computing resources efficiently to meet system requirements and maintaining that efficiency as demand changes (AWS, 2026).

Design Principles:

- Democratise advanced technologies (use managed services)
- Go global in minutes
- Use serverless architectures where possible
- Consider mechanical sympathy
- Experiment more often using on-demand resources

Application to NexaCart: NexaCart's on-premises T40 web servers are perpetually sized for peak traffic, wasting capacity during off-peak periods. EC2 Auto Scaling solves this by matching capacity to demand — scaling out from 2 to 4 instances during promotional campaigns and back down during quiet periods. Amazon RDS with read replicas can offload reporting queries from the primary database for vendor analytics dashboards.

1.8.5 Pillar 5: Cost Optimisation

Cost Optimisation focuses on avoiding unnecessary costs while achieving business requirements — understanding spending, choosing the right pricing models, and eliminating waste (AWS, 2026).

Design Principles:

- Adopt a consumption-based model
- Stop spending money on data centre operations
- Analyse and attribute expenditure
- Use managed services to reduce TCO

Application to NexaCart: The current on-premises infrastructure incurs fixed costs regardless of utilisation — idle servers during off-peak hours cost the same as peak periods. AWS's pay-as-you-go model, combined with Reserved Instances for baseline workloads and

Spot Instances for dev/test, can reduce NexaCart's compute costs by an estimated 40–60% over 3 years. AWS Cost Explorer and Budgets provide visibility to prevent cloud cost overruns.

1.8.6 Pillar 6: Sustainability

The Sustainability pillar focuses on environmental impacts, especially energy consumption and efficiency, since they are important levers for architects to inform direct action to reduce resource usage (AWS, 2026).

Design Principles:

- Understand your impact
- Establish sustainability goals
- Maximize utilization
- Anticipate and adopt new, more efficient hardware and software offerings
- Use managed services
- Reduce the downstream impact of your cloud workloads

Application to NexaCart: To enhance environmental sustainability, NexaCart can use auto-scaling technology to ensure compute is done only when there's a need (reducing wasted energy) and apply the right-sized RDS (Reducing wasted Energy). They can also reduce their footprint by utilizing S3 lifecycle policies to move older media to "colder," lower-energy consumption storage tiers and eliminating any obsolete/unknown data that is no longer necessary.

Business-wise, they can offer eco-friendly checkout methods like consolidated shipping or slow shipping/low emission deliveries and then measure orders/kWh as a main performance indicator for sustainability.

2. On-Premises Infrastructure Analysis & Migration Plan

2.1 Currently Existing Infrastructure

The existing on-premises infrastructure of NexaCart employs heterogeneous hardware from multiple vendors. There is a high level of operational complexity and increased risks associated with availability, scalability, and security.

2.1.1 Bottlenecks for Scalability

Asset	Specification	Limitation
Web Server (3 x Web Servers)	Dell Poweredge T-40, 16 GB RAM, 1 TB HDD;	Static Capacity: Web Servers do not scale to support web traffic spikes when there is a rush of shopper activity.
Application Server	HP ProLiant DL160, 32 GB RAM	Single port failure and cannot scale horizontally.
Database Server	Dell PowerEdge R640, 64 GB RAM	No replication for data: when you are doing synchronous writing operations, it will block reading data out.
File Server / NAS	Synology 10TB RAID	Can only access these files on the local server; no CDN to distribute media files.
Dev/Test Server	8 VCpu / 16 GB RAM - Virtualized	Has a significant amount of cost with no ability to scale as compared with production.

Table 1: Bottleneck for Scalability

The three web servers lack automated load balancing; a hardware load balancer would require manual configuration. During seasonal sales promotions, request volumes can spike 5–10 times the baseline, and the static server pool cannot absorb this demand elastically.

2.1.2 Availability & Performance Risk

- Single Application Server: The HP ProLiant DL160 running order management and business logic APIs is a single point of failure. Any hardware failure, planned maintenance, or OS patch triggers complete service downtime.
- Database with No Replication: The Dell PowerEdge R640 has no standby replica. A storage failure would require full restoration from the weekly backup rotation, potentially losing up to 7 days of order data.
- Backup System: Weekly rotation to external drives and NAS is inadequate; the RPO is approximately 7 days, unacceptable for a platform processing 60,000 monthly transactions.
- Network: Unmanaged 1 Gbps switches have no VLAN segmentation, meaning a compromised workstation has potential network access to the database server.

2.1.3 Security Risks

The Cisco ASA 5506-X firewall will give you the outer layer of protection against attacks; however, it does not protect against internal attacks via east-west traffic between your servers. Because there is no segmentation of the network (no control over how your PCI-Data is transmitted), you will not comply with PCI-DSS requirement 1 (network segmentation for cardholder data). In addition to this, the currently static IP based VPN (your only method of remote access) does not offer multi-factor authentication and there is no application firewall in place to prevent your e-commerce portal from being exposed to either SQL Injection or Cross-Site Scripting attacks.

2.1.4 Challenges from High-Traffic Events

During peak times such as the Colombo Sales Festival and Eid, NexaCart will experience tremendous traffic spikes that will exceed even your current web server's fixed capacity. Because of this, easily provisioning additional webservers is nearly impossible, resulting in poor customer experience. This problem directly results in lost revenue. There are many factors involved when adding a new physical server to an organization's network. Depending on those various factors, it usually takes multiple days (and often weeks) before a new physical server is fully operational on a business' network. Reactively managing your network's capacity in this manner is both impractical and inefficient.

2.2 Migration Strategy Design

Workload	Current Asset	Migration Strategy	Target AWS Service	Justification
Web Servers	3× Dell PowerEdge T40	Replatform	EC2 Auto Scaling Group (t3.micro/t3.small) behind ALB	Auto Scaling replaces static capacity with elastic provisioning.
Application Server	HP ProLiant DL160	Replatform	EC2 (t3.medium) behind ALB, App Tier	Managed OS patching, with the ability to scale horizontally.
Database Server	Dell PowerEdge R640	Replatform	Amazon RDS (MySQL/PostgreSQL) Multi-AZ	Automated failover, managed backups, read replicas.
File Server / NAS	Synology NAS, 10 TB	Replatform	Amazon S3 (MediaBucket) + CloudFront CDN	Unlimited scalability, 99.99% durability, global CDN
Email Server	MS Exchange 2019	Repurchase	Amazon WorkMail or Microsoft 365	Eliminates server maintenance; subscription model
Dev/Test Server	Virtualized 8 vCPU	Retire	EC2 Spot Instances (on demand)	No standing cost; launch on demand
Backup System	External drives + NAS	Replatform	AWS Backup + S3 Glacier	Automated, policy-driven

				backups; cross-region replication
Network Security	Cisco ASA 5506-X	Replatform	AWS Security Groups + WAF + Shield	Managed, scalable, and integrates with CloudTrail

Table 2: Migration Strategy Design

2.3 AWS Target Architecture Design

The target AWS architecture uses a 3-tier web app architecture based on the AWS Well-Architected Framework, running out of a Customer Virtual Private Cloud (VPC) that spans two different Availability Zones, thus giving you high availability.

2.3.1 Architecture Diagram (High-Level)

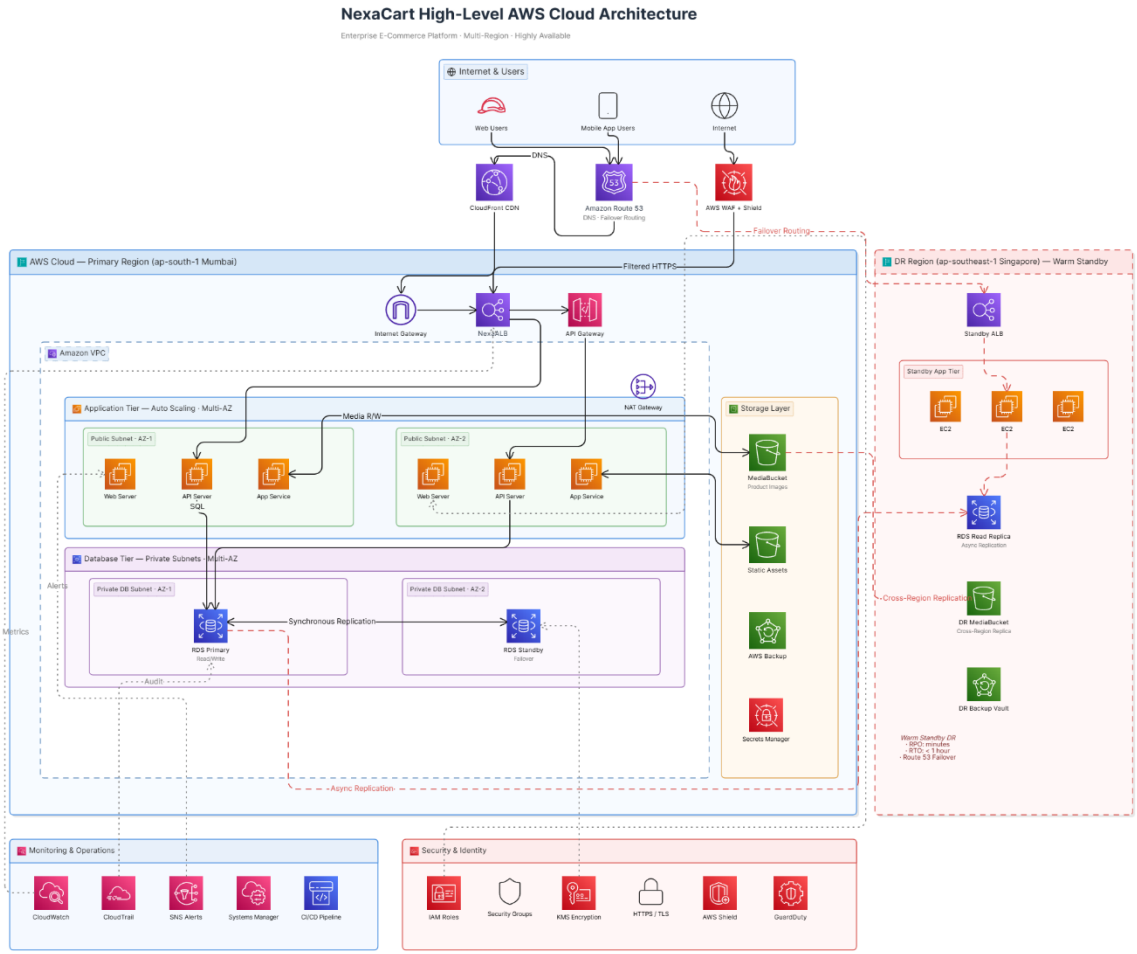


Figure 9: AWS Target High-Level Architectural Diagram

2.3.2 Architecture Diagram (Logical)

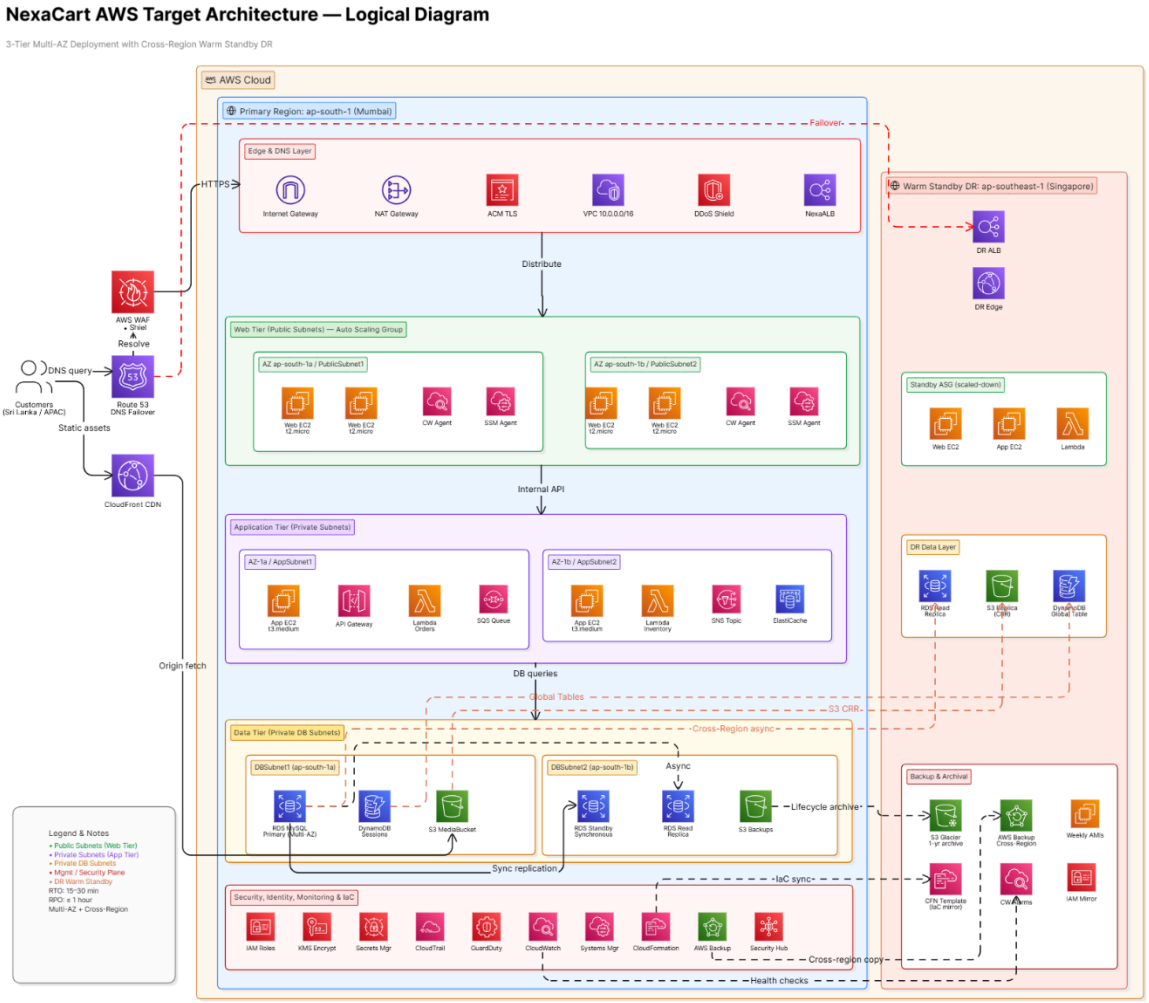


Figure 10: AWS Target Architectural Diagram

The above diagram is the full target architecture for NexaCart on AWS, including three tiers: the network tier (which includes Route53, internet gateway, application load balancer), compute tier (which includes an auto-scaling group of EC2 instances split out across 2 AZs, AppInstance) and data tier (which includes RDS multi-AZ with DBSubnet Group across the private subnets and S3 media bucket). The diagram also shows all security group boundaries as well as the flow of traffic through the various system components. The right side of the diagram shows the DR deployment of Warm Standby in ap-southeast-1 (Singapore) connected by asynchronous RDS replication and S3 Cross-Region Replication.

The key design decisions visible in the diagram are:

Multi-AZ compute- the Auto Scaling Group (min: 2, max: 4) places EC2 instances in both PublicSubnet1 (ap-south-1a) and PublicSubnet2 (ap-south-1b), eliminating any single-AZ SPOF at the web tier

- **Tiered security groups**- ALBSecurityGroup accepts inbound traffic from the internet; WebSecurityGroup accepts traffic only from the ALB; DBSecurityGroup accepts traffic only from the App/Web tier — enforcing least privilege access between tiers.
- **Private DB subnets**- NexaDB are deployed in DBSubnet1 and DBSubnet2, which have no direct internet route, consistent with the AWS Well-Architected Security pillar.
- **Route 53 DNS failover**- health checks on the primary ALB in Mumbai trigger automatic CNAME update to the Singapore ALB endpoint if the primary becomes unreachable, achieving the target RTO of 15–30 minutes.

3. CloudFormation Stack Design & Implementation

3.1 Infrastructure as Code

AWS CloudFormation defines, versions, and deploys the entire NexaCart infrastructure using code, eliminating any manual clicks in the console and assuring the reproducibility of the deployed infrastructure. The NexaCart-Advanced-ASG created on 14 April 2026 at 00:15:15 UTC+0530 in Region 'us-east-1' (N. Virginia) reached state 'CREATE_COMPLETE' for all 24 resources successfully provisioned.

3.2 Deployed CloudFormation Resources

The NexaCart-Advanced-ASG CloudFormation stack contains 24 resources covering network, compute, storage, database, and security layers:

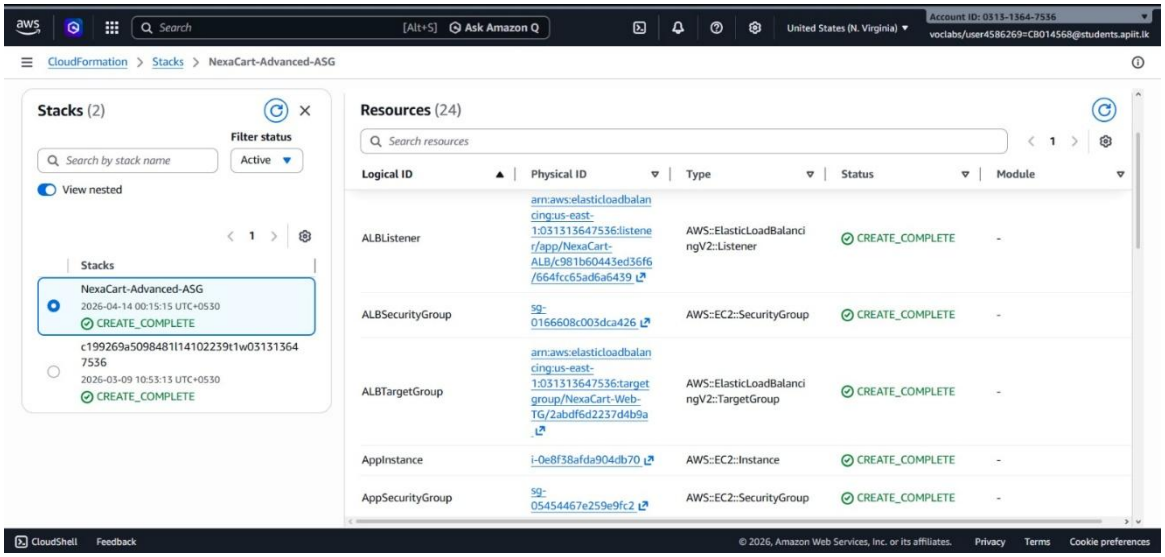


Figure 11: Screenshot of Deployed CloudFormation Resources Part 1

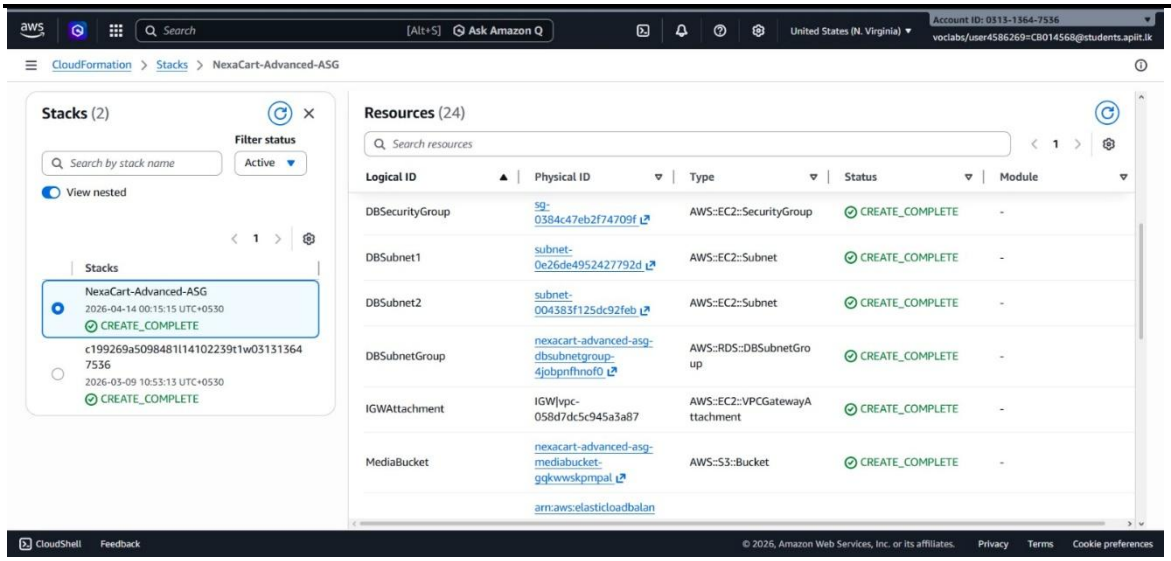


Figure 12: Screenshot of Deployed CloudFormation Resources Part 2

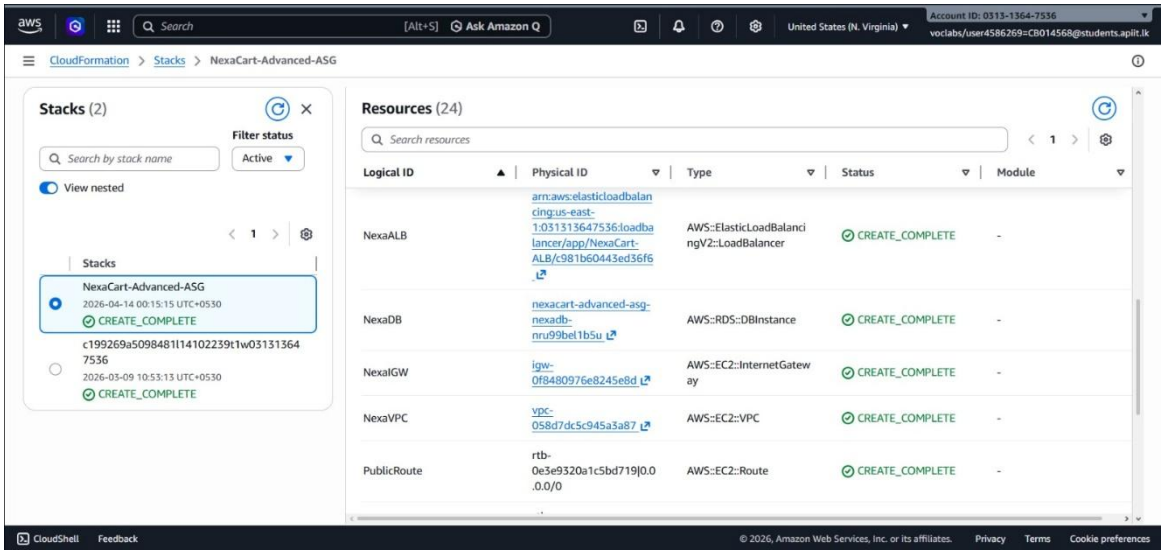


Figure 13: Screenshot of Deployed CloudFormation Resources Part 3

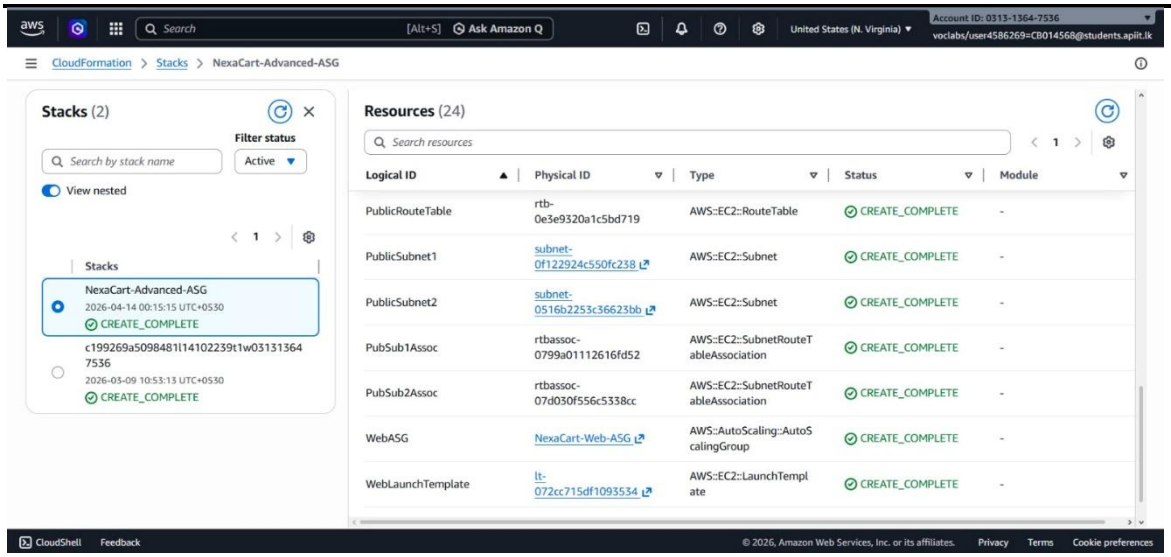


Figure 14: Screenshot of Deployed CloudFormation Resources Part 4

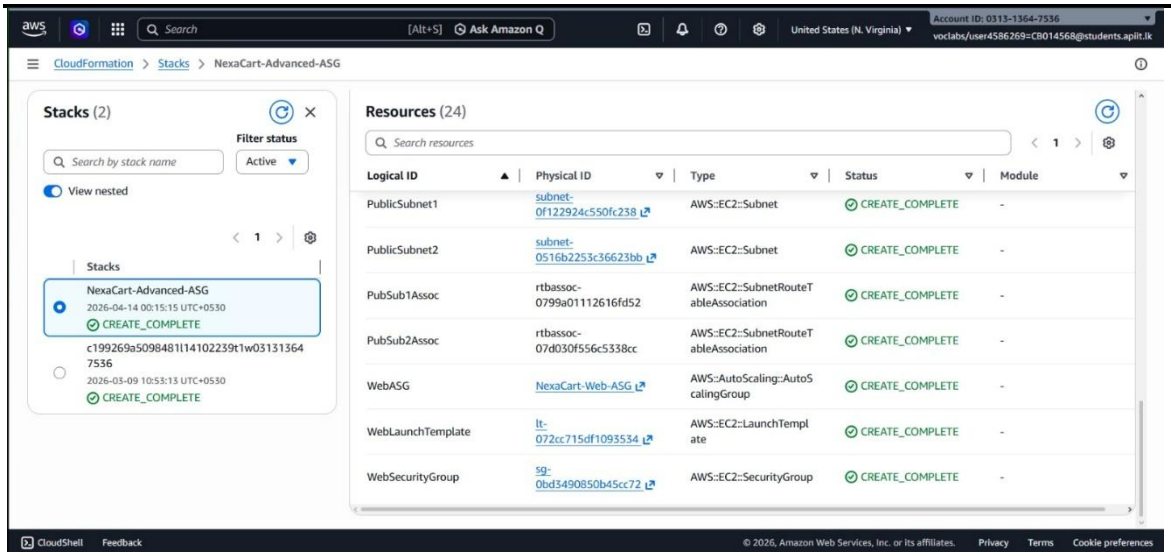


Figure 15: Screenshot of Deployed CloudFormation Resources Part 5

Logical ID	Physical ID	Type	Purpose	Component
NexaVPC	vpc-058d7dc5c945a3a87	AWS::EC2::VPC	Custom VPC	Virtual Private Cloud (VPC)
NexaIGW	igw-0f8480976e8245e8d	AWS::EC2::InternetGateway	Internet Access	Internet Gateway
IGWAttachment	IGW|vpc-058d7dc5945a3a87	AWS::EC2::VPCGatewayAttachment	Attaches IGW to VPC	Gateway Attachment
PublicRouteTable	rtb-0e3e9320a1c5bd719	AWS::EC2::RouteTable	Route table	Route table
PublicRoute	rtb-0e3e9320a1c5bd719|0.0.0.0/0	AWS::EC2::Route	Default route to IGW	Network Route
PublicSubnet1	subnet-0f122924c550fc238	AWS::EC2::Subnet	Web tier AZ-a	Subnet
PublicSubnet2	subnet-0516b2253c36623bb	AWS::EC2::Subnet	Web tier AZ-b	Subnet
PubSub1Assoc	rtassoc-0799a01112616fd52	AWS::EC2::SubnetRouteTableAssociation	Associate subnet1 to the route table	Route Table Association
PubSub2Assoc	rtassoc-07d030f556c5338cc	AWS::EC2::SubnetRouteTableAssociation	Associate subnet2 to the route table	Route Table Association
DBSubnet1	subnet-0e26de4952427792d	AWS::EC2::Subnet	DB tier AZ-a	Subnet

DBSubnet2	subnet-004383f125dc92feb	AWS::EC2::Subnet	DB tier AZ-b	Subnet
DBSubnetGroup	nexacart-advanced-asg-dbsubnetgroup-4jobpnfhnof0	AWS::RDS::DBSubnetGroup	RDS multi-AZ subnet group	Database Subnet Group
NexaDB	nexacart-advanced-asg-nexadb-nru99bel1b5u	AWS::RDS::DBInstance	Production database	Relational Database (RDS)
NexaALB	arn:aws:elasticloadbalancing:us-east-1:031313647536:loadbalancer/app/NexaCart-ALB/c981b60443ed36f6	AWS::ElasticLoadBalancingV2::LoadBalancer	Application Load Balancer	Load Balancer
ALBListener	arn:aws:elasticloadbalancing:us-east-1:031313647536/listener/app/NexaCart-ALB/c981b60443ed36f6/664fcc65ad6a6439	AWS::ElasticLoadBalancingV2::Listener	HTTP port 80 listener	Load Balancer Listener

ALBTarget Group	arn:aws:elasticloadbalancing:us-east-1:031313647536./targetgroup/NexaCart-Web-TG/2abdf6d2237d4b9a	AWS::ElasticLoadBalancingV2::TargetGroup	EC2 health check targets	Target Group
WebLaunch Template	lt-072cc715df1093534 (NexaCartWebLT)	AWS::EC2::LaunchTemplate	EC2 instance config	Launch Template
WebASG	NexaCart-Web-ASG	AWS::AutoScaling::AutoScalingGroup	Auto Scaling Group	Auto Scaling Group (ASG)
MediaBucket	nexacart-advanced-asg-mediabucket-gqkwwskpmpal	AWS::S3::Bucket	Media/file storage	S3 Storage Bucket
ALBSecurityGroup	sg-0166608c003dca426	AWS::EC2::SecurityGroup	ALB inbound rules	Security Group (Firewall)
WebSecurityGroup	sg-0bd3490850b45cc72	AWS::EC2::SecurityGroup	Web tier inbound rules	Security Group (Firewall)
AppSecurityGroup	sg-05454467e259e9fc2	AWS::EC2::SecurityGroup	App tier rules	Security Group (Firewall)

DBSecurity Group	sg-0384c47eb2f74709f	AWS::EC2::Security Group	DB tier inbound rules	Security Group (Firewall)
AppInstance	i-0e8f38afda904db70	AWS::EC2::Instance	Application server	EC2 Virtual Server

Table 3: Deployed CloudFormation Resources

3.3 Launch Template Configuration

The NexaCartWebLT defines the EC2 instance configuration for the Auto Scaling Group:

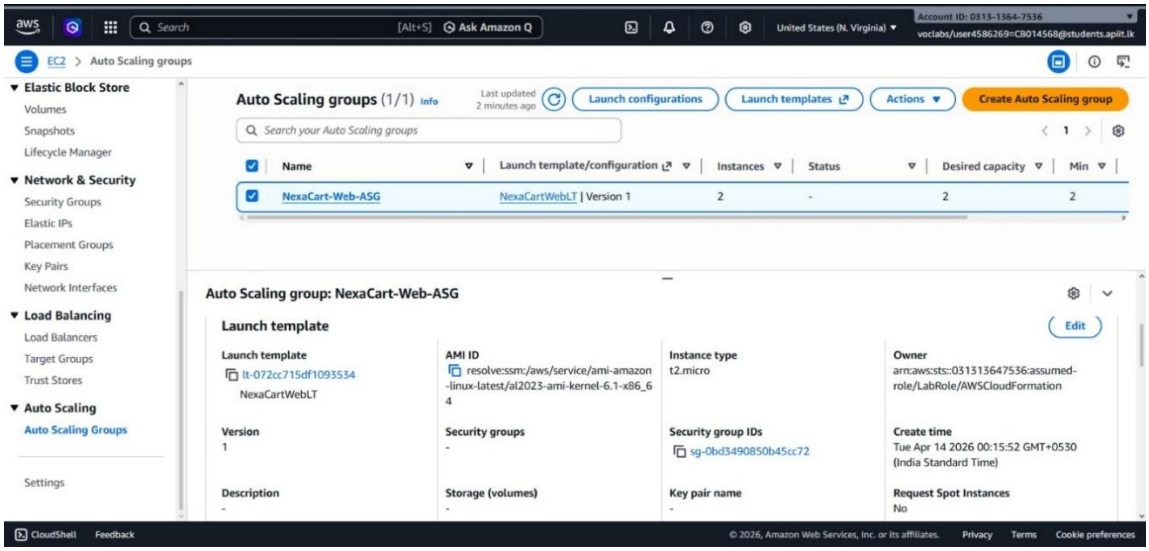


Figure 16: Screenshot of the Launch Template Configuration

3.4 Auto Scaling Group Configuration

The NexaCart-Web-ASG provides elastic compute capacity:

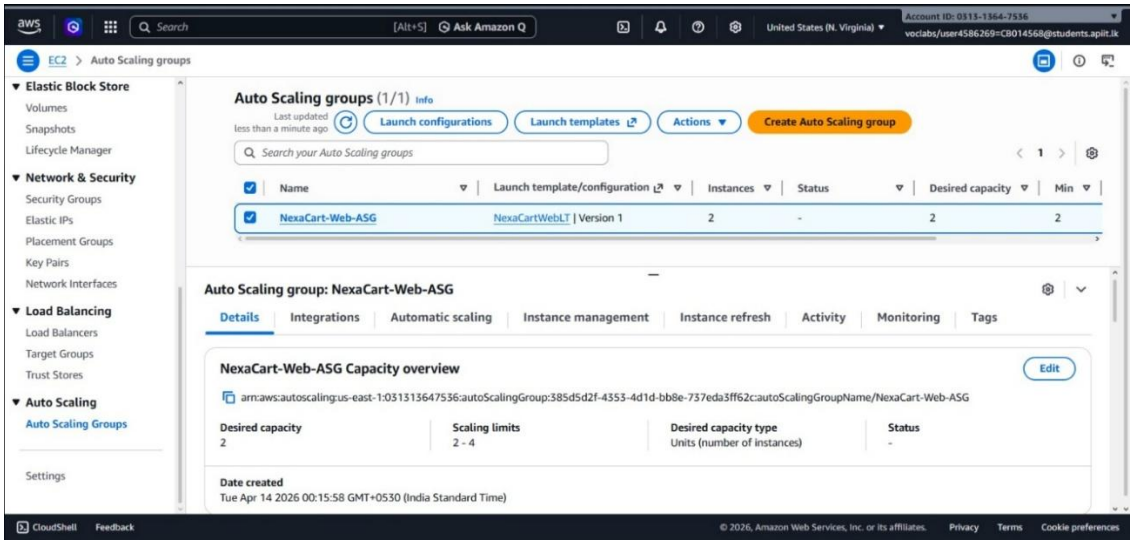


Figure 17: Screenshot of Auto Scaling Group Configuration

3.5 Live Web Server



Figure 18: Screenshot of Live website

3.6 CloudFormation Best Practices Applied

The NexaCart-Advanced-ASG stack follows AWS CloudFormation best practices:

- Dynamic AMI References: Using SSM Parameter Store to resolve the latest Amazon Linux 2023 AMI avoids hardcoding AMI IDs that become stale across regions.
- No Credentials in Templates: Sensitive values (DB passwords) should be stored in AWS Secrets Manager and referenced via {{resolve:secretsmanager:...}} dynamic references.
- Modular Security Groups: Separate security groups per tier (ALB, Web, App, DB) implement least-privilege network access - each tier can only receive traffic from the immediately upstream tier.
- IaC as Single Source of Truth: All resources defined in CloudFormation prevent configuration drift - manual changes outside the stack would create inconsistency.
- Parameterised Stack: Using CloudFormation parameters for instance types, capacity values, and VPC CIDR blocks enables the same template to deploy identical infrastructure across dev/staging/production environments.

3.7 AWS Academy Lab Limitations

Certain enterprise-grade resources could not be deployed within the AWS Academy Lab environment due to service restrictions:

- HTTPS ALB Listener (port 443): Requires an ACM public certificate and a registered domain name — not feasible in the lab environment. In production, SSL/TLS termination at the ALB using a free ACM certificate would be implemented.
- CloudFront Distribution: Content Delivery Network in front of S3 MediaBucket for low-latency global asset delivery was not deployed due to lab limitations.
- AWS WAF: Web Application Firewall integration with NexaALB for OWASP Top 10 protection.
- Multi-AZ RDS: The NexaDB instance was provisioned as a single-AZ instance to manage lab resource constraints; production deployment would enable Multi-AZ with automated failover.
- VPC Endpoints: Private connectivity to S3 and DynamoDB via VPC endpoints to avoid traversing the public internet.

4. AWS Disaster Recovery Strategy

4.1 DR Strategy Overview

AWS defines four primary disaster recovery strategies, each offering different trade-offs between cost, complexity, RTO, and RPO:

Strategy	RTO	RPO	Cost	Complexity	Description
Backup & restore	Hours	Hours	Lowest	Low	Periodic snapshots; restore from backup on failure.
Pilot Light	Minutes -1 hour	Minutes -1 hour	Low - Medium	Medium	Core data replicated; minimal services running.
Warm Standby	Minutes	Seconds - minutes	Medium - High	High	Scaled-down functional copy; scale up on failover.
Multi-Site Active-Active	Seconds	Near - Zero	Highest	Very High	Full capacity in multiple regions simultaneously.

Table 4: Current DR Strategy Overview

4.2 Recommended Strategy: Warm Standby

For NexaCart Solutions, the Warm Standby strategy is recommended — providing the optimal balance between recovery speed, cost, and complexity given NexaCart's operational scale (~60,000 monthly orders, ~150 employees) and business requirements around payment processing availability.

Justification:

RTO Target: 15–30 minutes NexaCart processes an average of ~2,000 orders per day (~83 per hour). A 1–2-hour outage during peak trading could result in significant revenue loss and damage to customer trust. The Warm Standby approach achieves 15 - 30-minute RTO through pre-provisioned infrastructure in a secondary region that can be scaled up rapidly. This is materially better than Backup & Restore (hours) but more cost-effective than Multi-Site Active-Active.

RPO Target: ≤ 1 hour with Amazon RDS automated backups (daily snapshots + continuous transaction log archival to S3), NexaCart can achieve an RPO of approximately 5 minutes

using point-in-time recovery. For the secondary region, RDS Read Replicas with asynchronous replication typically achieve RPO of seconds to minutes.

Cost Justification: The Warm Standby maintains a scaled-down environment in a secondary AWS region (e.g., ap-south-1 Mumbai, geographically close to Sri Lanka). The secondary environment runs minimal instances (1 web instance instead of 2, smaller RDS instance class) that can be scaled up to full production capacity within minutes of a failover declaration. This is significantly less expensive than Multi-Site Active-Active while offering far better recovery times than Backup & Restore.

4.3 DR Architecture Components

NexaCart's customer base is in Sri Lanka and wider South Asia. Routing production traffic to us-east-1 (N. Virginia) introduces round-trip latency of 200–250 ms^{-1} from Colombo, which is unacceptable for an e-commerce checkout flow where sub-100 ms^{-1} latency directly impacts conversion rates. The correct primary region is ap-south-1 (Mumbai, India), which delivers approximately 30–50 ms^{-1} round-trip from Colombo — roughly a 5–7× latency improvement. The warm standby region is ap-southeast-1 (Singapore), which is geographically close (≈3,750 km from Mumbai), has excellent network interconnectivity within the AWS APAC backbone, and introduces only ~60 ms^{-1} additional latency from Sri Lanka in a failover scenario.

Justification: Selecting a region offers a balance of user experience and resiliency against disasters. Mumbai has the lowest latency amongst all regions considered and hence provides optimal conditions for achieving high conversion rates for e-commerce. Singapore was selected as the secondary region since it has sufficient distance (geographically) otherwise to prevent a localized natural disaster or other geopolitical event from affecting India but remains close enough to India to allow asynchronous data replication to occur with low latency.

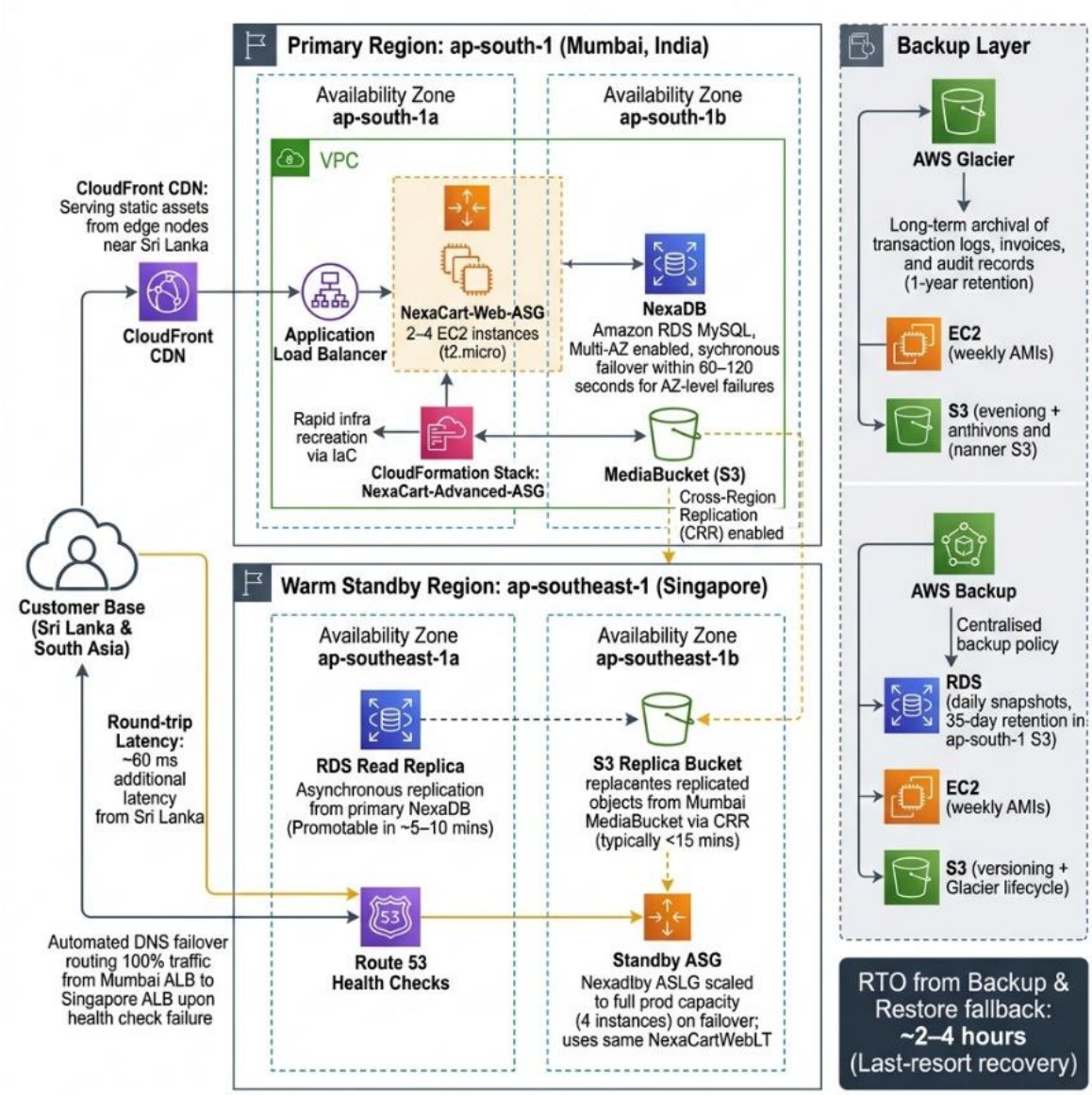


Figure 19: NexaCart AWS architecture

4.3.1 Primary Region Architecture (ap-south-1 – Mumbai)

The main region contains the working production instance of the application and is built for high availability by deploying into multiple Availability Zones.

4.3.1.1 Computer Layer

The application's deployment will be hosted using an Auto Scaling Group (ASG), which will consist of at least 2 (to 4) EC2 (t2.micro) instances distributed across 2 Availability Zones (AZ).

To distribute incoming request traffic across all EC2 instances in ASG, I will use Application Load Balancer (ALB). This allows us to scale (add or remove) the resources as needed and provide fault tolerance for your production environment.

Justification: Utilizing an ASG across two AZs prevents a single data center outage from taking down the application (ensuring High Availability). The t2.micro instances provide a highly cost-effective baseline computing layer that can dynamically scale during traffic spikes (e.g., flash sales). The ALB ensures traffic is evenly distributed, performs health checks to route traffic away from failing instances, and offloads SSL/TLS termination to reduce the compute burden on the EC2 instances.

4.3.1.2 Database Layer

The system uses Amazon RDS MySQL with Multi-AZ deployment, which provides:

- Synchronous data replication.
- Automatic failover within 60–120 seconds.
- High availability during Availability Zone failures.

Justification: Transaction Data Loss: E-commerce Apps Can't Afford to Lose Transactions (dropped orders). Multi-AZ Synchronous Replication Provides Zero RPO for Localized AZ Failures. Automated Failover Mechanism Prevents Human Intervention for an AZ-level Database Crash and Protects Revenue Streams 24x7%.

4.3.1.3 Storage Layer

An AWS S3 bucket (MediaBucket) is used to store static files (product images).

To support disaster recovery:

- Cross-Region Replication (CRR) is enabled
- Data is automatically replicated to the Singapore region

Justification: Decoupling static media from the EC2 compute layer drastically reduces EBS storage costs and server load. CRR ensures that if the Mumbai region goes down, the Singapore standby environment will already have a synchronized copy of all product images, preventing broken media links upon failover.

4.3.1.4 Infrastructure Management

The infrastructure is defined using **AWS CloudFormation**, enabling:

- Infrastructure as Code (IaC)
- Rapid recreation of environments

- Reduced Recovery Time Objective (RTO)

Justification: In the event of an emergency, manual configuration carries a substantially higher likelihood of human error, as well as a considerably more stressful situation for the person deploying the resources. The use of CloudFormation will also ensure that there is always a consistent configuration between the DR and Primary environments. Therefore, if the Primary facility experiences a total failure, the infrastructure can be audited, updated, and recreated within a matter of minutes, resulting in dramatic reductions in RTO.

4.3.1.5 Content Delivery

A **CloudFront CDN** is configured with the S3 bucket as its origin. This allows:

- Static content delivery from edge locations near Sri Lanka
- Reduced latency and improved user experience

Justification: E-commerce sites are highly visual. CloudFront caches heavy assets (product images, CSS, JavaScript) at edge locations closest to the users. This not only further reduces perceived load times for Sri Lankan customers but also significantly reduces the egress bandwidth and request load hitting the origin S3 bucket.

4.3.2 Warm Standby Region (ap-souteast-1 Singapore)

The Singapore region acts as a **disaster recovery (DR) site**, operating in a warm standby configuration.

4.3.2.1 Database Replication

An **RDS Read Replica** is maintained using asynchronous replication from the primary database.

- Can be promoted to primary within **5-10 minutes**
- Ensures minimal data loss

Justification: Cross-region synchronous replication is physically impossible without introducing massive latency to the primary database. Asynchronous replication provides a near real-time backup. By maintaining an active Read Replica, the DR site avoids a cold boot of the database, enabling a rapid promotion to Primary status to meet the strict 5-10-minute RTO.

4.3.2.2 Storage Replication

The **S3 Replica Bucket** receives data via CRR from Mumbai.

- Replication delay: typically, less than 15 minutes
- Ensures data consistency across regions

Justification: A 15-minute replication delay is highly acceptable for static assets (product photos, UI elements), as these do not represent mission-critical transactional data (like an order checkout).

4.3.2.3 Compute Readiness

A **standby Auto Scaling Group** is maintained with minimal capacity (0–1 instances).

- Scales up to full production capacity (4 instances) during failover
- Use the same launch template as the primary region

Justification: Operating at 0-1 instances keeps DR operational costs to an absolute bare minimum during normal operations. By relying on the ASG and shared launch templates, the system can autonomously expand to meet production traffic demands the moment Route 53 directs traffic to Singapore.

4.3.2.4 Failover Mechanism

Route 53 Health Checks continuously monitor the primary region.

- On failure detection, DNS automatically reroutes traffic
- 100% traffic shifts from Mumbai to Singapore

Justification: DNS-level routing is decoupled from regional AWS outages. Route 53's globally distributed nature ensures that the failover trigger remains highly available even if the entire Mumbai AWS region is offline, ensuring a seamless, automated transition for users.

4.3.3 Backup & Archival Strategy

A centralized backup solution is implemented using AWS Backup:

- **RDS:** Daily snapshots with 35-day retention
- **EC2:** Weekly AMIs
- **S3:** Versioning enabled

For long-term compliance:

- **S3 Glacier** is used to archive logs, invoices, and audit records
- Data retention period: 1 year

Justification: AWS Backup provides a unified, automated policy rather than relying on fragmented, service-by-service backup scripts. The 35-day RDS retention covers end-of-

month reconciliation periods for e-commerce accounting. S3 Versioning acts as an immediate safety net against accidental overwrites or ransomware. S3 Glacier is chosen for 1-year compliance retention because its ultra-low-cost tier is perfectly suited for "write-once, read-rarely" financial and audit data.

4.3.4 Recovery Objectives

Failure Scenario	Recovery Time
Availability Zone failure	60-120 seconds
Region failover (Singapore)	~5-10 minutes
Backup & restore fallback	~2-4 hours

Table 5: Recovery Objectives

4.4 DR Runbook (Warm Standby Failover Procedure)

1. **Detection (0–5 min):** CloudWatch alarms in ap-south-1 detect ALB unhealthy target count = 0 or RDS instance unavailability; Route 53 health checks begin failing.
2. **Assessment (5–10 min):** On-call engineer confirms region-level failure in Mumbai (not AZ-level, which RDS Multi-AZ handles automatically within 60–120 seconds without human intervention)
3. **Database Failover (10–15 min):** Promote RDS Read Replica in ap-southeast-1 (Singapore) to standalone primary; promotion completes in ~5–10 minutes
4. **Compute Scale-Up (15–20 min):** Update Singapore standby ASG desired capacity from 1 to 4; instances launch from pre-defined Launch Template in ap-southeast-1
5. **DNS Cutover (20–30 min):** Route 53 weighted routing policy automatically directs 100% traffic to Singapore ALB endpoint (or manual override if auto-failover not triggered)
6. **Validation (25–35 min):** Smoke test order placement, payment processing, and inventory queries on Singapore environment
7. **Total RTO: ~25–35 minutes**

4.5 DR Cost vs. Complexity Trade-offs

Factor	Backup & restore	Warm Standby (Recommended)	Active-Active
Monthly Estimated Cost	~$50-100 additional	~$200-400 additional	~$800-1,200 additional
RTO	2-4 hours	15-30 minutes	< 5 minutes
RPO	1-24 hours	5-60 minutes	Near-zero
Maintenance Effort	Low	Medium	High
Suitable for NexaCart	Minimum baseline	**Recommended**	Future Scale
Primary Region	ap-south-1 (Mumbai)	ap-south-1 (Mumbai)	ap-south-1 + ap-southeast-1
Secondary Region	None	ap-southeast-1 (Singapore)	ap-southeast-1 (Singapore)

Table 6: DR Cost vs. Complexity Trade-offs

The Warm Standby Method is best suited for NexaCart's current size because it offers significant improvements in RTO/RPO compared to standard backup/restore strategies and provides these improvements at much lower costs when compared with a full active-active deployment. Both primary and warm standby (in Singapore) serve Sri Lanka while maintaining sub-100 ms round trip latencies, allowing end users to understand no loss of performance, even during a DR failover.

5. Conclusion

Through its move to AWS, NexaCart Solutions has gone from having an unreliable, manually operated, limited capacity physical location to an efficient, automated and elastic cloud environment. The company used the 7Rs framework to guide the decisions associated with each individual workload migration and to support the company's overall strategy for AWS. NexaCart has replatformed its core services on AWS managed solutions such as RDS, S3, ALB; repurchased legacy email; and retired dev/test infrastructure that was not in use. The five pillars of the AWS Well-Architected Framework provide the quality bar for how NexaCart's architecture will operate, ensuring it is operationally effective, secure, reliable, and most importantly, performance-efficient while being cost-effective. The CloudFormation implementation (NexaCart-Advanced-ASG, 24 resources, CREATE_COMPLETE) demonstrates Infrastructure as Code maturity, enabling reproducible deployments and rapid disaster recovery. NexaCart's on-premises limitations can be resolved by using an Auto Scaling Group (min:2, max:4) to handle peak-traffic scalability issues. The Warm Standby DR strategy will improve the existing backup system's RTO (15–30 minutes) and RPO (≤1 hour) target values from the previous 7 days backup system to meet reducing RTO/RPO target values. With these decisions regarding the architecture, NexaCart is well-positioned to grow throughout South Asia, roll out order tracking in real time, and implement demand forecasting using artificial intelligence on a platform that supports scalability, security, and dependability. The primary geographic deployment of NexaCart will be ap-south-1 (Mumbai) and secondary (warm standby) will be ap-southeast-1 (Singapore). This configuration will allow for optimal production performance (sub-50ms latency from Colombo) and disaster recovery capabilities to serve NexaCart's APAC customers.

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